Preventive Medicine 163 (2022) 107196

Contents lists available at ScienceDirect

Preventive Medicine

journal homepage: www.elsevier.com/locate/ypmed

The association of substance use with attaining employment among unemployed job seeking adults: Prospective findings from the French CONSTANCES cohort

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ARTICLE INFO

Keywords:
Tobacco use
Alcohol use
Cannabis use
Employment

ABSTRACT

This study aimed to examine the prospective association between tobacco, alcohol and cannabis use with attaining employment among unemployed job seekers. Data from the French population-based CONSTANCES cohort on 5114 unemployed job seeking adults enrolled from 2012 to 2018 were analyzed. Binary logistic regressions were computed. Odds ratio (OR) and 95%CI of remaining unemployed at one-year of follow-up (versus attaining employment) according to substance use at baseline were obtained. The following independent variables were introduced into separate models: tobacco use (non-smoker, former smoker, light (<10cig/day), moderate (10-19cig/day) and heavy smoker (>19cig/day)), alcohol use according to the Alcohol Use Disorder Identification Test (non-users (0), low (<7), moderate (7–15) and high or very high-risk (>15)) and cannabis use (never used, no use in the previous 12 months, less than once a month, at least once a month but less than once per week, once per week or more). Analyses were adjusted for age, gender and education. At follow-up, 2490 participants (49.7%) were still unemployed. Compared to non-smokers, moderate and heavy smokers were more likely to remain unemployed, with ORs (95%CI) of 1.33 (1.08–1.64) and 1.42 (1.04–1.93), respectively. Compared to low-risk alcohol users, no alcohol users and high or very high-risk alcohol users were more likely to remain unemployed, with ORs (95% CI) of 1.40 (1.03–1.83) and 2.10 (1.53–2.87), respectively. Compared to participants who never used cannabis, participants who use cannabis once a week or more were more likely to remain unemployed, OR (95%CI) of 1.63 (1.33–2.01). Substance use may play an important role in difficulty attaining employment.

1. Introduction

Substance use is one of the most important risk factors for adult mortality and morbidity in the world (Griswold et al., 2018). Besides many adverse consequences on physical and mental health, substance use is also negatively linked to educational achievements and positively with poverty and social deprivation (Kuntsche et al., 2004; Semple, 2015). From a public health perspective, these detrimental consequences are driven by the most frequently consumed substances: tobacco, alcohol and cannabis (Griswold et al., 2018; Marten et al., 2018; World Drug Report 2021 [Internet], 2021). France has one of the highest rates of tobacco and cannabis use among Western European countries,

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https://doi.org/10.1016/j.ypmed.2022.107196
Received 10 January 2022; Received in revised form 28 July 2022; Accepted 7 August 2022
Available online 9 August 2022
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and in 2017, at least 23% of the French general population exceeded the recommended maximum amounts for alcohol consumption (*Consommation d'alcool en France: où en sont les Français ? [Internet]*, 2021; European Monitoring Centre for Drugs and Drug Addiction, 2021; *Statistical Bulletin* 2021 — Prevalence of Drug Use [Internet], 2021). On the other hand, according to the International Labor Organization, unemployment is a worldwide crisis aggravated by the COVID-19 pandemic and its rate is expected to rise to 5.7% in 2022 (*Slow Jobs Recovery and Increased Inequality Risk Long-Term COVID-19 Scarring [Internet]*, 2021). In France, unemployment rate was roughly stable in the 10 years prior to the Covid-19 global crisis, i.e., from 8.9% in 2010 to 7.8% in 2020 (*Au quatrième trimestre 2020, le taux de chômage se replie à nouveau, à 8,0% - Informations rapides - 037 | Insee [Internet]*, 2022). Age, gender, and education remained among the top sociodemographic factors associated with employment. Specifically, an older age, being a woman and being less educated are frequently positively associated with an increased likelihood of unemployment (Wanberg et al., 2016; Andersson, 2015; Henkel, 2011). Unemployment has been associated with physical and mental health concerns and with an increased risk of morbidity and mortality (Bambra, 2010; Sorlie and Rogot, 1990). Stress reactions, anxiety and depressive symptoms were shown to be common responses to job loss (Guindon and Smith, 2002). In addition, unemployment increases the risk of poverty, contributes to inequality and may jeopardize access to health care (Bambra, 2010; Huang et al., 2014). Moreover, it has been linked to unhealthy behaviors (Morris et al., 1992), such as greater substance use (i.e., tobacco, alcohol and cannabis use). In observational studies, unemployed individuals had a higher prevalence of substance use compared to those with a job (Henkel, 2011; Compton et al., 2014), and substance use was found to significantly increase as a response to job loss (Morris et al., 1992; Deb et al., 2011). In addition, substance use increases the likelihood of losing one's job (Airagnes et al., 2019). Although the adverse consequences of substance use on job loss and reciprocally have been thoroughly examined, the role of substance use on employment among job seeking individuals has not been well studied (Prochaska et al., 2016; Michalek et al., 2020; Claussen, 1999; Skärlund et al., 2012).

A longitudinal observational study in California examined the differences between 120 non-smokers in the previous year before study enrolment and 131 smokers; in attaining employment over a 12-month period. The results showed that non-smokers were 24% more likely to be reemployed compared to smokers after adjustment for sociodemographic and clinical factors (Prochaska et al., 2016).

Few studies investigated the role of alcohol use in employment with conflicted findings (Claussen, 1999; Skärlund et al., 2012; Jørgensen et al., 2019). Claussen et al. found that at-risk alcohol use (Alcohol Use Disorder Identification Test (AUDIT) >10) was significantly more prevalent among unemployed than reemployed participants at five-years of follow-up (Skärlund et al., 2012). While, Skärlund et al. reported that at-risk alcohol use (AUDIT >8) was not significantly associated with unemployment at one-year of follow-up (Skärlund et al., 2012). Several studies reported that chronic illicit drug use, including cannabis use, was associated with reduced likelihood of employment (DeSimone, 2002; French et al., 2001; Alexandre and French, 2004). However, to the best of our knowledge, even though, there is literature documenting the association of unemployment and tobacco use, data on cannabis and alcohol use is not yet well studied using longitudinal data. Therefore, the aim of this study was to examine prospectively the association of tobacco, alcohol and cannabis use with attaining employment at one-year of follow-up among job seeking adults at baseline while considering sociodemographic and clinical factors and duration of the last unemployment period. We took advantage of the French national population-based CONSTANCES cohort that includes a large sample of adults from various sociodemographic backgrounds and are followed annually. We hypothesized that individuals who use substances would be less likely to get a job and that this association would follow different pathways for each substance. Our findings may be informative for preventive approaches targeting individuals who are seeking a job.

2. Methodology

2.1. Participants

The French population-based CONSTANCES cohort randomly recruited participants aged 18 to 69 at inclusion between 2012 and 2020 (n = 202,674 in 2020). At baseline and then annually, participants completed self-administered questionnaires that assessed health-related behaviors, occupational conditions and sociodemographic factors (Zins and Goldberg, 2015). The CONSTANCES Cohort was authorized by the French Data Protection Authority (Commission Nationale de l'Informatique et des Libertés, CNIL) and approved by the Institutional Review Board of the National Institute for Medical Research – INSERM (no. 01–011). All the participants provided an informed consent.

In the present study, the sample was restricted to individuals who had a previous work experience but were unemployed and seeking a job upon study enrolment (n = 11,333). From the latter sample, participants who had data on employment or unemployment at one year of follow-up were selected and included in the statistical analyses (n = 5114). Participants who were included after 2018 (n = 955) who had no follow-up data (n = 4005), reported being a student, retired, not working for health reasons, or other (n = 1078) and did not report their employment status (n = 181) were excluded from the analyses (Fig. 1).

2.2. Substance use at baseline

2.2.1. Tobacco

Smoking status was self-reported at baseline and categorized as follows: never smoker, former smoker or current smoker. Among current smokers, the number of cigarettes per day was further collected. Tobacco use was then categorized as follows: 1) Never smokers, 2) Former smokers, 3) Current light smokers (1–9 cigarettes per day), 4) Current moderate smokers (10–19 cigarettes per day), and 5) Current heavy smokers (>19 cigarettes per day), as defined in several previous studies (Lohse et al., 2016; Clift et al., 2022).

2.2.2. Alcohol

Alcohol use at baseline was evaluated using the French version of the Alcohol Use Disorders Identification Test (AUDIT) that includes 10 items (Gache et al., 2005). The total AUDIT score was obtained by adding the scores of the ten items and was then categorized as such: 1) No use (a score of 0), 2) Low-risk (1–7), 3) Moderate risk (8–15), 4) High or very high-risk (>15).

2.2.3. Cannabis

At baseline, the frequency of cannabis consumption was self-reported by answering the following questions: 1) "Have you ever consumed cannabis? If yes, over the past 12 months, have you consumed cannabis? And 2) Over the past 30 days, have you consumed cannabis? If yes, how many times have you consumed cannabis over the past 30 days?". We computed a categorical variable from the previous questions as such: 1) Never consumed 2) No consumption during the last 12 months; 3) At least once during the last 12 months but less than once a month; 4) At least once a month but less than once per week; and 5) Once per week or more.

2.2.4. Employment status at follow-up

Employment status was self-reported at one-year of follow-up with the following question: "What is your current employment situation?": 1) employed, including on sick leave, leave without pay or availability, maternity/paternity/adoption/parental leave; 2) unemployed and job seeker; 3) retired or withdrawn from business; 4) in training (high school student, student, trainee, apprentice or other; 5) does not work for

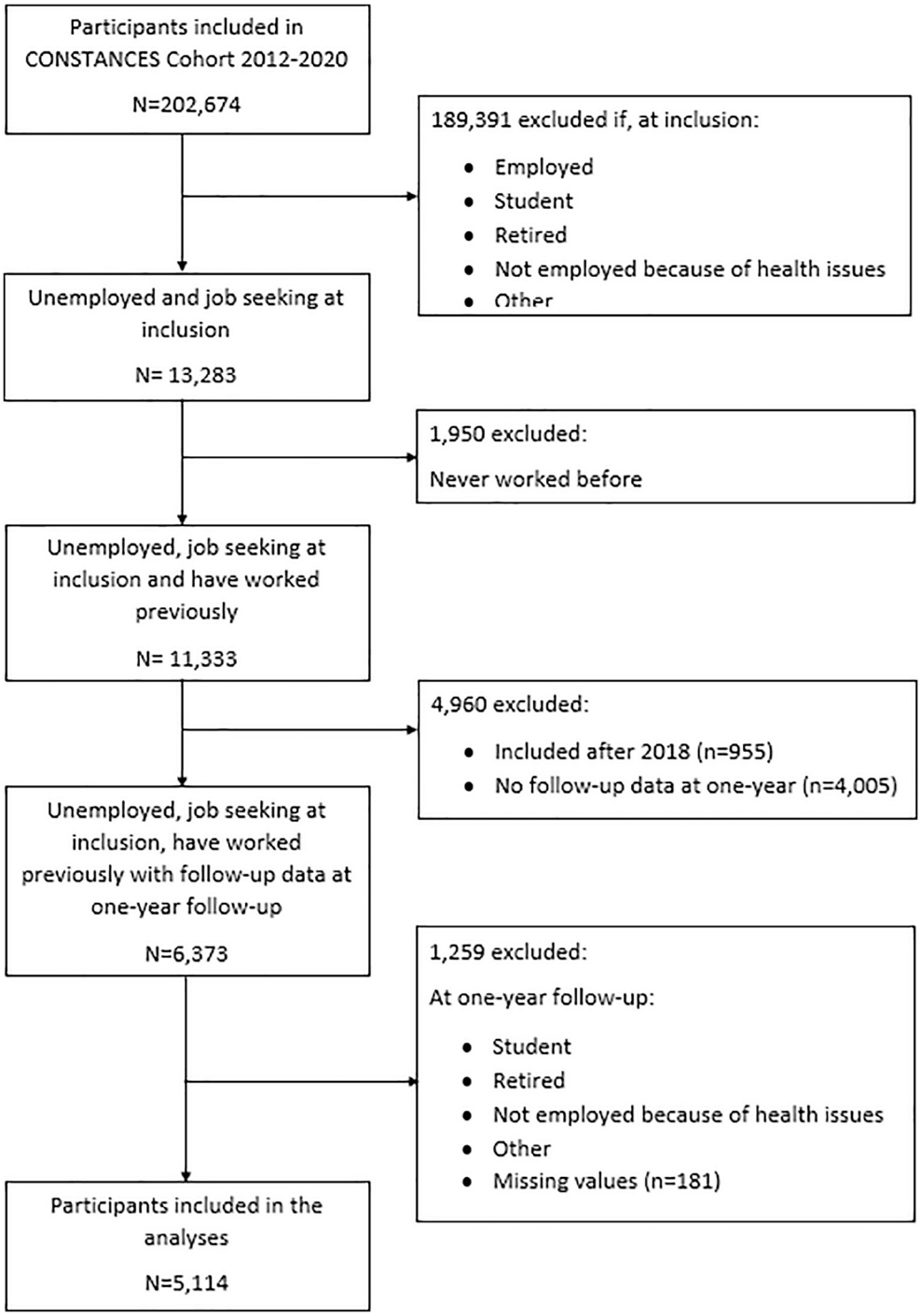

Fig. 1. Flow chart of study population selection.

health reasons; 6) at home without employment; 7) other situation. In the present analyses, we only included participants who answered "employed" or "unemployed and seeking a job".

2.2.5. Covariates at baseline

We considered the following sociodemographic variables at baseline: age, gender and education. Age was analyzed as a continuous variable. Education was based on the International Standard Classification of Education (ISCED 2011) and modeled as a categorical variable as follows: less than an associate degree (level 0–4), associate degree or more (level 5–8) (International Standard Classification of Education (ISCED) [Internet], 2021).

The duration of unemployment was considered as a binary variable: unemployed for one year or less and unemployed for more than a year (*Définition - Chômeur de longue durée | Insee [Internet]*, 2021).

At baseline, depressive symptoms were evaluated with the Center of Epidemiologic Studies Depression scale (CESD). The CESD score was dichotomized and a score ≥ 19 was considered as indicating a depressive state (sensitivity/specificity for the diagnosis of major depression: 0.85/0.86) (Morin et al., 2011).

Self-rated health was reported at baseline with the following question: "How would you describe your general health compared to someone you know of the same age?". Participants could answer from 1 = "Very good" to 8 = "Very Poor".

2.3. Statistical analysis

Descriptive statistics for the prevalence of substance use were conducted according to employment status at one-year of follow-up and chi-square or *t*-tests were calculated as adequate (i.e., categorical versus continuous variables). Second, logistic regression analyses were conducted separately for the association of each substance use at baseline and the outcome of not attaining employment at one year of follow-up. The analyses were adjusted for age, gender and education.

Third, several supplementary analyses were performed:

1. The analyses were stratified by the duration of unemployment at baseline to examine whether the role of substance use may differ according to the duration of unemployment.
2. The analyses were re-conducted after the exclusion of participants with a depressive state at baseline.
3. The three substances were introduced simultaneously in the same model to examine if one substance has a stronger association with employment rate compared to others.
4. The analyses were rerun with education as a continuous variable to examine if the associations could differ while taking into account the five categories of ISCED 2011 (assuming that these categories are ordinal representation of underlying sets of continuous units such as years of education). (Winship and Mare, 1984).
5. Interactions between the date of the year of follow-up and the substances were tested to account for yearly fluctuations.
6. Analyses excluding participants with poor self-reported health at baseline (score > 5) were conducted because poor health condition may interfere in the search process and in finding an employment.
7. *E*-values were calculated to estimate the minimal estimate of an association that an unmeasured confounder would need to have with both the employment status and the substances to fully explain the reported associations (VanderWeele and Ding, 2017).

The following variables had missing data: education (1.4%), AUDIT score (18.4%), tobacco use (8%) duration of unemployment (18%) and depressive status (4%) which was handled by multiple imputation with 10 datasets (Haukoos and Newgard, 2007).

A two-sided value of $P < 0.05$ was considered statistically significant. All analyses were conducted using SPSS IBM Statistics for Windows version 21.

3. Results

3.1. Participants' characteristics

At baseline, 64.4% of the 5114 participants had been unemployed for less than a year. From the 5114 participants, 4.4% were current heavy smokers, 4.1% were alcohol users at high or very high-risk, and 10.1% consumed cannabis at least once per week. At one-year of follow-up, 2490 participants (49.7%) were still unemployed. Participants' characteristics according to employment status at follow-up are presented in Table 1 and all the bivariate associations between substances and employment status at follow-up were significant. Characteristics of the 11,333 job seekers at baseline according to their employment status at one year of follow-up are presented in Supplemental Table 1.

3.2. Tobacco use

Current moderate and heavy smokers had increased odds of not getting an employment at one-year of follow-up compared to non-smokers: ORs [95% CI] 1.33 [1.08–1.64] and 1.42 [1.04–1.93], respectively (Table 2).

Table 1
Characteristics of the 5114 participants included at baseline according to their employment status at one- year of follow-up.

	Total	Employed	Unemployed	Chi square	*P* value
	5114 (100%)	2624 (51.3)	2490 (48.7)		
Tobacco use				44.82	<0.001
Non-smoker	2110 (41.3)	1169 (44.5)	941 (37.8)		
Former smoker	1617 (31.6)	776 (29.6)	841 (33.8)		
Current light smoker	591 (11.6)	327 (12.5)	264 (10.6)		
Current moderate smoker	570 (11.1)	264 (10.1)	306 (12.3)		
Current heavy smoker	226 (4.4)	88 (3.3)	138 (5.5)		
Alcohol use (AUDIT)[a]				26.54	<0.001
No use (0)	230 (4.5)	105 (4.0)	125 (5.0)		
Low risk (1–7)	3799 (74.3)	1999 (76.2)	1799 (72.2)		
Moderate risk (8–15)	876 (17.1)	446 (17)	431 (17.3)		
High or very high-risk (>15)	209 (4.1)	74 (2.8)	135 (5.4)		
Cannabis consumption				43.42	<0.001
Never consumed	2427 (47.5)	1201 (45.8)	1226 (49.2)		
Not during the previous year	1655 (32.4)	892 (34.0)	763 (30.6)		
At least once during the last 12 months but less than once a month	329 (6.4)	207 (7.9)	122 (4.9)		
At least once a month but less than once per week	189 (3.7)	106 (4)	83 (3.3)		
Once per week or more	514 (10.1)	218 (8.3)	296 (11.9)		
Age	41.98 ± 11.52	38.7 ± 10.65	45.35 ± 11.35	−21.27[e]	<0.001
Gender				19.86	<0.001
Men	2147 (42.0)	1023 (39.0)	1124 (45.0)		
Women	2967 (58.0)	1601 (61.0)	1366 (54.9)		
Education[b]				84.07	<0.001
Less than an associate degree	2219 (43.4)	976 (37.2)	1243 (49.9)		
Associate degree or more	2895 (56.6)	1648 (62.8)	1247 (50.1)		
Duration of unemployment				203.18	<0.001
< 1 year	3293 (64.4)	1934 (73.7)	1360 (54.6)		
≥ 1 year	1821 (35.6)	690 (26.3)	1130 (45.4)		
Depressive Symptoms[c]				27.89	<0.001
No	3745 (73.2)	2009 (76.6)	1736 (69.7)		
Yes	1369 (26.8)	615 (23.4)	754 (30.3)		
Self-rated health Score[d]				25.50	<0.001
≤ 5	4684 (91.6)	2455 (93.6)	2229 (89.5)		
> 5	430 (8.4)	169 (6.4)	229 (10.5)		

[a] Alcohol use disorders identification test to evaluate alcohol use.
[b] Based on the international standard classification of education.
[c] Measured using the Center of Epidemiologic Studies Depression scale and a score ≥ 19.
[d] Self-rated health score from 1 = "very good" to 8 = "very poor".
[e] Independent *t*-test for continuous variable.

Table 2
Association between tobacco use and not returning to employment at one-year follow-up ($n = 5114$).

Not employed at one-year of follow-up	OR	95% CI	
Tobacco use			
Non-smoker	Ref	–	–
Former smoker	1.06	0.91	1.22
Current light smoker	1.11	0.91	1.36
Current moderate smoker	**1.33**	**1.08**	**1.64**
Current heavy smoker	**1.42**	**1.04**	**1.93**
Age	**1.05**	**1.04**	**1.06**
Gender			
Men	Ref	–	–
Women	**0.88**	**0.78**	**0.99**
Education[a]			
Less than an associate degree	Ref	–	–
Associate degree or more	**0.72**	**0.64**	**0.81**

[a] Based on the international standard classification of education.

3.3. Alcohol use

Participants who do not consume alcohol and those who are at-high or very high-risk of consumption at inclusion had a higher odd of not getting employed at one-year of follow-up compared to low-risk users: 1.40 [1.03–1.83] and 2.10 [1.53–2.87], respectively (Table 3). In addition, the odds of not getting employed in participants with a high or very high-risk alcohol use was significantly higher than in non-users (*Z*-score = 3.32, $p < 0.001$).

3.4. Cannabis use

Consuming cannabis at least once per week was associated with an increased odd of not getting an employment at one-year of follow-up 1.63 [1.33–2.01] (Table 4).

3.5. Supplementary analyses

1) After stratifying for the duration of unemployment at baseline (one year or less versus more than one year), the only associations that remained significant were for participants who have been unemployed for one year or less (Supplemental Table 2). There were no significant interactions however between substances and the duration of unemployment (all p for interactions >0.05).
2) The exclusion of participants with a depressive state at baseline ($n = 1369$), did not change the interpretation of the results (Supplemental Tables 3–5).
3) The addition of the three substances in the same model yielded the same interpretations for participants who do not use alcohol (OR 1.47, 95%CI 1.07 to 2.00), for high or very high-risk alcohol users (OR 1.88, 95%CI 1.37 to 2.60) and for cannabis use at least once per week (OR 1.50, 95%CI 1.19 to 1.88). However, the associations with tobacco use did not persist (Supplemental Table 6).
4) Using education as a continuous variable did not change the significance of the results, except for no alcohol use, for which effect size the was similar but did not reach significance ($p = 0.05$) (Supplemental Tables 7–9).
5) There were no interactions between the number of years of follow-up and the three substances (all $p > 0.137$).
6) After excluding participants with a poor self-rated health at baseline, all associations remained significant with a similar effect sizes, except for heavy tobacco use (Supplemental Tables 10–12).
7) Unmeasured confounders would have to have an OR of 1.57-fold increased risk of unemployment and be 1.57 times more prevalent in moderate tobacco use to explain the observed odds ratio for moderate smoking (95%CI 1.24 – Not applicable (NA)). These e-values were of 1.67 (95%CI 1.16 - NA) for heavy tobacco use, 1.64 (95%CI 1.13 - NA) for no alcohol use, 2.25 (95%CI 1.77 - NA) for at high or very high-risk alcohol use and 1.87 (95%CI 1.57 - NA) for at least weekly cannabis use.

Table 3
Association between alcohol use and not returning to employment at one-year of follow-up ($n = 5114$).

Not employed at one-year of follow-up	OR	95% CI	
Alcohol use (AUDIT)[a]			
Low risk	Ref	–	–
No use	**1.40**	**1.03**	**1.89**
Moderate risk	1.17	0.99	1.38
High or very high-risk	**2.10**	**1.53**	**2.87**
Age	**1.05**	**1.04**	**1.05**
Gender			
Men	Ref	–	–
Women	0.91	0.81	1.03
Education[b]			
Less than an associate degree	Ref	–	–
Associate degree or more	**0.72**	**0.64**	**0.81**

[a] Alcohol use disorders identification test to evaluate alcohol use.
[b] Based on the international standard classification of education.

Table 4
Association between cannabis consumption and not returning to employment at one-year of follow-up ($n = 5114$).

Not employed at one-year of follow-up	OR	95% CI	
Cannabis consumption			
Never consumed	Ref	–	–
Not during the previous year	1.06	0.93	1.21
At least once during the last 12 months but less than once a month	0.92	0.71	1.18
At least once a month but less than once per week	1.25	0.92	1.70
Once per week or more	**1.63**	**1.33**	**2.01**
Age	**1.05**	**1.04**	**1.06**
Gender			
Men	Ref	–	–
Women	0.90	0.80	1.00
Education[a]			
Less than an associate degree	Ref	–	–
Associate degree or more	**0.71**	**0.63**	**0.80**

[a] Based on the international standard classification of education.

4. Discussion

In the present study, we aimed to examine prospectively the association between tobacco, alcohol and cannabis use with attaining employment at one-year of follow-up among job seeking unemployed adults from a national population-based cohort. All the three substances were significantly associated with employment status at one-year of follow up after adjusting for age, gender and education. Moderate or heavy smoking, no alcohol use or using alcohol at high or very high-risk and at consuming cannabis at least once per week were associated with decreased likelihood of attaining employment at follow-up. All these associations remained significant after the exclusion of participants with a depressive state at baseline. These associations concern especially unemployed job seekers for one year or less and the role of alcohol and cannabis use were stronger than tobacco use.

Current moderate and heavy smokers were less likely to be employed at follow-up. This result supports the findings of an earlier longitudinal study which found that non-smokers were more likely to be employed compared to smokers but where no comparison between never smokers and former smokers was made (Prochaska et al., 2016). A meta-analysis on smoking and employment concluded that smokers were 33% more likely to be absent from work and to take extra sick leave compared to non-smokers (Weng et al., 2013) and a study by Berman et al. estimated that in the US, a smoking employee costs an extra $5816 annually (Berman et al., 2014). These issues could discourage employers from recruiting smokers. In fact in a qualitative study, unemployed job seeking daily smokers expressed their concern that smoking prevented them from finding a job and mentioned their feeling about the physical characteristics of smoking (e.g., the smell of cigarette smoke, altered vocal chords and teeth discoloration) that could have impeded their chances of getting employed (Michalek et al., 2020). Furthermore, on study showed that cigarette consumption ranks second behind food about spending behaviors among job seeking heavy smokers (Stinson et al., 2020). Thus, smoking was prioritized over job seeking resources (e.g., cell phone, transportation) and health care.

Alcohol non-users were more likely to remain unemployed at follow-up compared to low-risk alcohol users while high or very high-risk alcohol users had a higher likelihood of staying unemployed. In line with previous studies regarding health outcomes, these results reveal a non-linear relationship between alcohol use and employment (Claussen, 1999; Jørgensen et al., 2019; Johansson et al., 2007). Low-risk alcohol users may have better physical and mental health than those who are heavy drinkers and individuals who do not consume alcohol at all (because some people do not drink for health reasons and some are also former heavy drinkers who have stopped drinking (Stockwell et al., 2016). However, since not using alcohol remained associated with employment status after the exclusion of participants with poor self-rated health, other factors may account for these associations. For instance, since alcohol consumption is a common habit in France, (Hill, 2021) no alcohol consumption may limit social networking and therefore opportunities to meet potential employers or increase the chance to hear more about new job offers. Moreover, high-risk alcohol use can promote cognitive disorders (e.g., motivation processes), inappropriate behaviors (e.g., disinhibition, irritability) and health issues (e.g., hepatitis, pancreatitis) leading to decreased chances of finding a job. In addition, employers may notice alcohol consumption during job interviews (e.g., smell of alcohol or withdrawal symptoms such as tremors).

Participants who use cannabis for at least once a week were more likely to remain unemployed at follow-up compared to participants who do not use it. This result is in line with prior literature regarding the decreased employment rate in middle-aged adults who were regular cannabis users during their adolescence, compared to non-users or experimenters (Windle and Wiesner, 2004). Regular cannabis use can adversely affect cognitive functioning (e.g., difficulty in attention and concentration, deterioration in thought structuring and expression, memory disorders), psychomotor functioning (e.g., perceptual and motor coordination), physiological functioning (e.g., drowsiness, sleep disorders) and can promote psychiatric disorders (Abuse NI on D, 2021). Such symptoms may hinder job searching, performance during job interviews which may decrease the likelihood of getting a job. In addition, at least in France where cannabis is illegal, employers may be reluctant to hire cannabis users, whenever noticeable, in order to avoid legal issues.

Our study is, to the best of our knowledge, the first to assess the association between tobacco, alcohol and cannabis use and the likelihood of getting employed among unemployed adults who are seeking a job. Moreover, we adjusted for sociodemographic factors and we took into consideration the duration of unemployment and the depressive state using a standardized assessment tool.

This study has several limitations. First, although the participants of the CONSTANCES cohort were randomly recruited, voluntary participation is subject to selection bias and thus the participants are not representative of the general population. They have higher education level and consume less tobacco, alcohol or cannabis compared to the general population (Fry et al., 2017). Second, since data were collected before the Covid-19 pandemic, findings must be extrapolated with great caution to the pandemic period. Third, response rate at follow-up was of 59.7% and non-responders could differ from responders such as their use of substances which was higher at baseline for individuals who did not respond at follow-up. It must be noted however that participants who did not respond to follow-up had not answered the entire questionnaire and not specifically the question about their employment status. Unfortunately, this loss to follow-up can lead to a decreased statistical power. Fourth, since the data on substance use were self-reported, they could be subject to social desirability and/or recall bias. However, self-reported health risk behaviors via confidential questionnaires was found to be one of the collection methods that were least associated with social desirability bias (Crutzen and Göritz, 2010; Link and Mokdad, 2005). Fifth, we cannot draw a causal conclusion since it is an observational study and unmeasured confounding factors may play a role in the associations between substance use and employment (e.g., personality traits including impulsivity, environments with few opportunities, being part of a sexual minority among other factors). However, education, a variable known to be strongly associated with both substance use and employment (Beard et al., 2021; Sata et al., 2021) had a strength of association of 1.4. Thus, according to the calculated e-values (1.57 for moderate smoking, 1.67 for heavy smoking, 1.64 for no alcohol use, 2.25 for high or very high alcohol use and 1.87 for at least weekly cannabis use), such confounders should have a greater effect size than education to fully explain the reported associations, and even an effect size of at least 60% larger than that of education for high or very high alcohol use (Chamorro et al., 2012; Hart, 2017).

5. Conclusion

Our results suggest that public health campaigns should inform the general population (i.e., employed and unemployed subjects) on the greatest difficulties of substance users for being employed when they seek a job. Such information might be useful to motivate individuals who have started looking for a job to try decreasing their substance use and seek help before they get into a vicious circle of a worsening of substance use and prolonged unemployment. The associations between substance use and employment were only significant among participants who have been unemployed for a year or less. This highlights the urge for considering substance use at the very beginning of unemployment since after a prolonged period of unemployment (e.g., more than a year), other factors may take over the role of substance use in finding a job such as discouragement leading to a less active job search, loss of professional networks and reluctance of employers to hire a long-term unemployed person. Further studies should investigate the benefits of the SBIRT approach (Screening, Brief Intervention, Referral to Treatment) on the odds of employment among job seekers. Lastly, the role of other substances (e.g., stimulants, benzodiazepines, hallucinogens) should be investigated (Wamsley et al., 2018).

Funding

The Constances Cohort benefits from a grant from ANR (ANR-11-INBS-0002). Constances is also partly funded by MSD and L'Oréal. None of these funding sources had any role in the design of the study, collection and analysis of data or decision to publish.

Authors' contributions

Rita El Haddad and Guillaume Airagnes conceived and designed the analysis. Rita El Haddad performed the analysis and wrote the

manuscript. Cédric Lemogne contributed as well to the methodology. Rita El Haddad, Guillaume Airagnes, Cédric Lemogne, Joane Matta, Emmanuel Wiernik, Marcel Goldberg, Maria Melchior, Yves Roquelaure, Frédéric Limosin, and Marie Zins contributed to the study design, interpretation of the findings and critically revised the manuscript. All authors approved the final version of the paper.

Declaration of Competing Interest

Guillaume Airagnes has received speakers and/or consulting fees from Pfizer, Lundbeck and Pierre Fabre. Cédric Lemogne has received speakers and/or consulting fees from Lundbeck, Boehringer Ingelheim, Janssen and Otsuka Pharmaceutical. However, these entities did not have any role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript. Rita El Haddad, Joane Matta, Emmanuel Wiernik, Marcel Goldberg, Maria Melchior, Yves Roquelaure, Frédéric Limosin, and Marie Zins report no conflicts of interest.

Data availability

The data that has been used is confidential.

Acknowledgments

The authors thank the "Population-based Epidemiologic Cohorts Unit" (Cohortes épidémiologiques en population), Université de Paris Cité, INSERM, Paris Saclay University, UVSQ, UMS 011, which designed and manages the Constances Cohort Study. They also thank the National Health Insurance Fund ("Caisse nationale d'assurance maladie", CNAM) and its Health Screening Centres ("Centres d'examens de santé"), which are collecting a large part of the data, as well as the National Old-Age Insurance Fund (Caisse nationale d'assurance vieillesse) for its contribution to the constitution of the cohort, ClinSearch, Asqualab and Eurocell, which are conducting the data quality control.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ypmed.2022.107196.

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